

PricePilot AI

Dynamic Pricing Optimization & Revenue Intelligence System

Milestone-3 Documentation

Competitor Analysis, Revenue Optimization,
Business Intelligence & Frontend Implementation

Submitted By
Aditya Raj Pandey

Project Objective

The objective of PricePilot AI is to provide an AI-powered dynamic pricing and revenue intelligence platform that helps businesses make data-driven pricing decisions using demand patterns, product information, inventory conditions, competitor pricing, promotions and historical sales data.

Milestone 3 extends the Machine Learning, backend and database foundation established in Milestone 2 into a complete business-facing application. The focus is on competitor analysis, market intelligence, revenue optimization, profitability analytics, executive business intelligence and frontend implementation.

Major Objectives of Milestone 3

- Implement competitor monitoring and pricing comparison workflows.
- Develop market intelligence and competitive positioning interfaces.
- Build revenue optimization and profitability analysis workflows.
- Implement pricing simulation and scenario-based analysis.
- Generate pricing strategy recommendations.
- Develop executive business intelligence dashboards.
- Develop and integrate the complete React frontend.
- Implement interactive charts, tables and KPI components.
- Generate structured business reports.
- Integrate the AI assistant with the application interface.
- Provide responsive and consistent user experiences.

Milestone Context

Milestone 2 established the ML models, FastAPI application layer, PostgreSQL database, authentication, product, inventory, competitor, pricing, sales and AI services. Milestone 3 focuses on presenting and connecting these capabilities through business workflows.

Step 1: Frontend Environment Setup

The frontend development environment was configured using a modern React and Vite stack. The frontend is separated from the backend while communicating with it through API services.

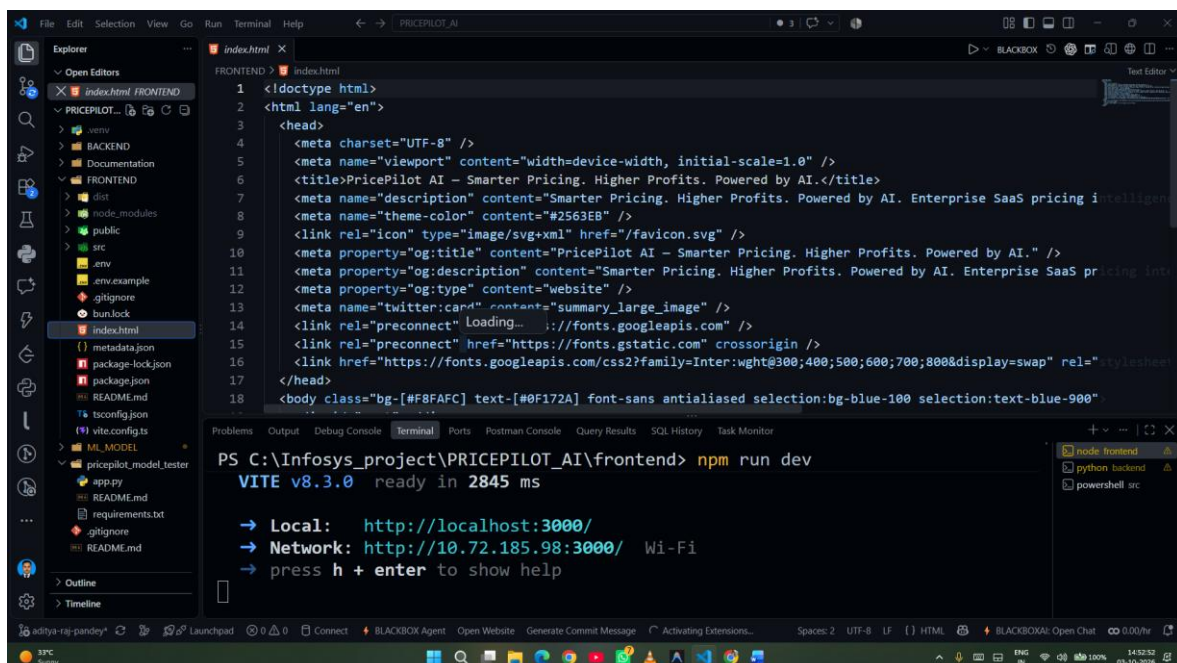
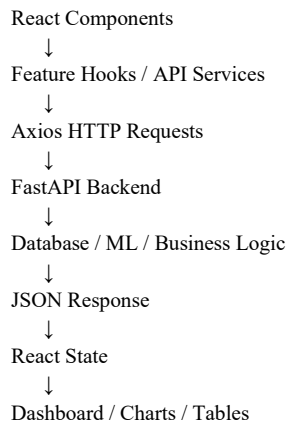
Technologies Used

- React
- Vite
- JavaScript
- Tailwind CSS
- React Router
- Axios
- Recharts
- Lucide React
- Motion

Frontend Development Goals

- Create a professional enterprise SaaS interface.
- Provide clear navigation between business modules.
- Use reusable components and feature-based organization.
- Connect frontend workflows with backend APIs.
- Display organization data rather than hardcoded business results.
- Provide loading, empty and error states.
- Maintain responsive behavior across desktop, tablet and mobile.

Development Flow



Step 2: Frontend Architecture & Navigation

A feature-based frontend architecture was used to separate global UI elements from business-specific modules. This approach improves maintainability and makes it easier to extend individual business domains.

Frontend Structure



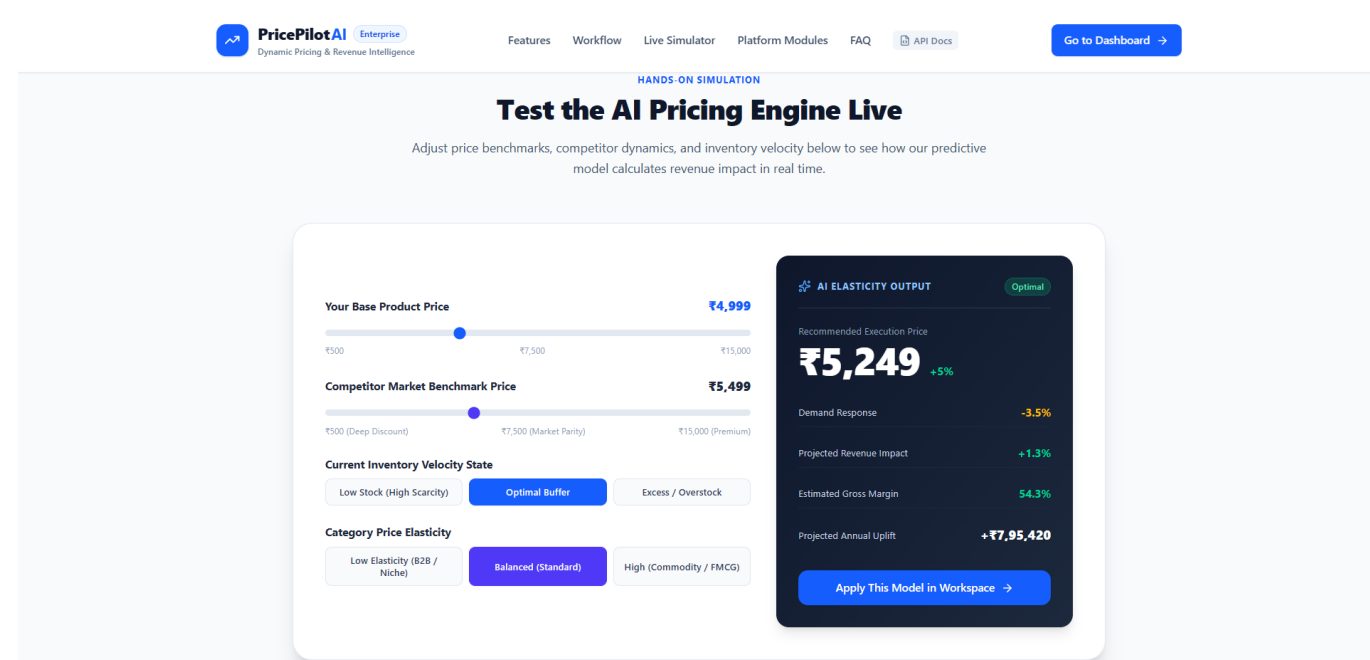
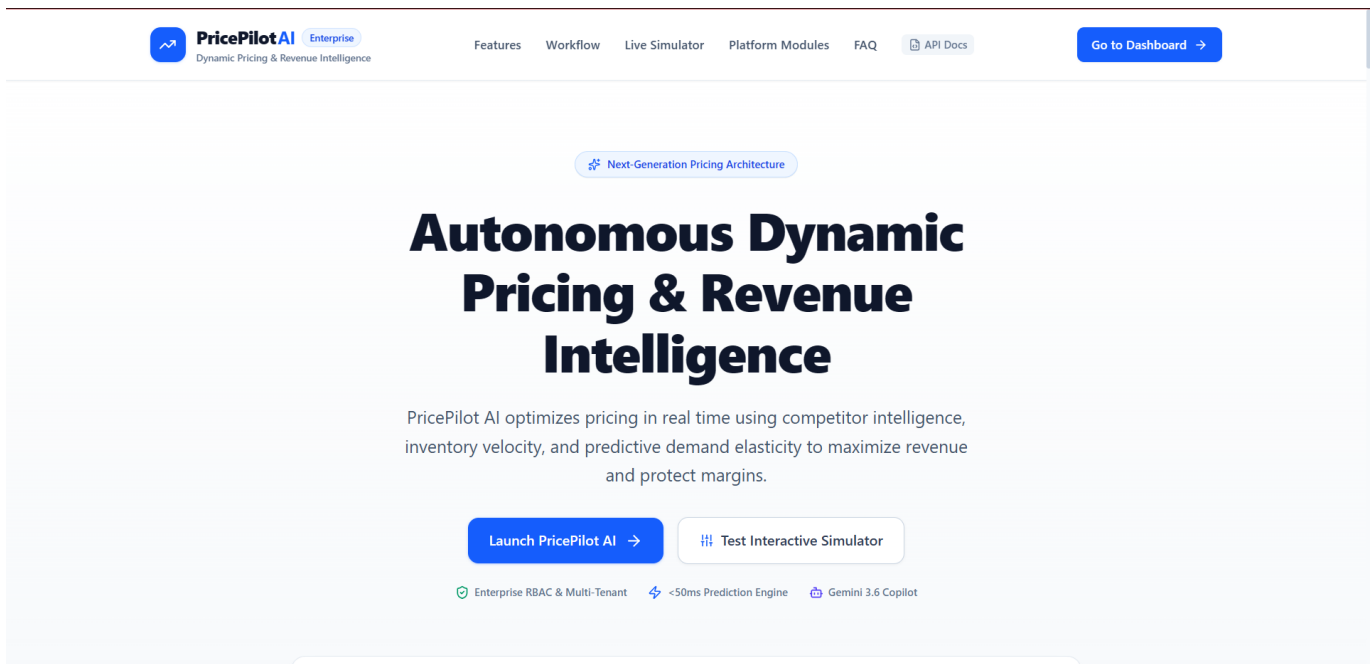
Application Navigation

- Landing Page
- Login / Registration
- Dashboard
- Products
- Categories
- Inventory
- Competitors
- Pricing
- Forecasting
- Revenue Optimization
- Pricing Analytics
- Profitability
- Market Intelligence

- Reports
- Executive Business Intelligence

Protected Navigation

Unauthenticated users enter through the public landing page and authentication screens. Business modules are presented as protected application routes after authentication.



Step 3: Authentication, Dashboard & Core Management

Authentication Frontend

Dedicated Login and Registration interfaces were implemented. The frontend manages form states, validation, loading states and API errors while the backend remains responsible for authentication and authorization.

- Login interface
- Registration interface
- Authentication state handling
- Protected routes
- Error and loading states

Main Dashboard

The main dashboard acts as the central business intelligence interface. It presents high-level KPIs and visual summaries for revenue, demand, pricing, inventory, competitors and product performance.

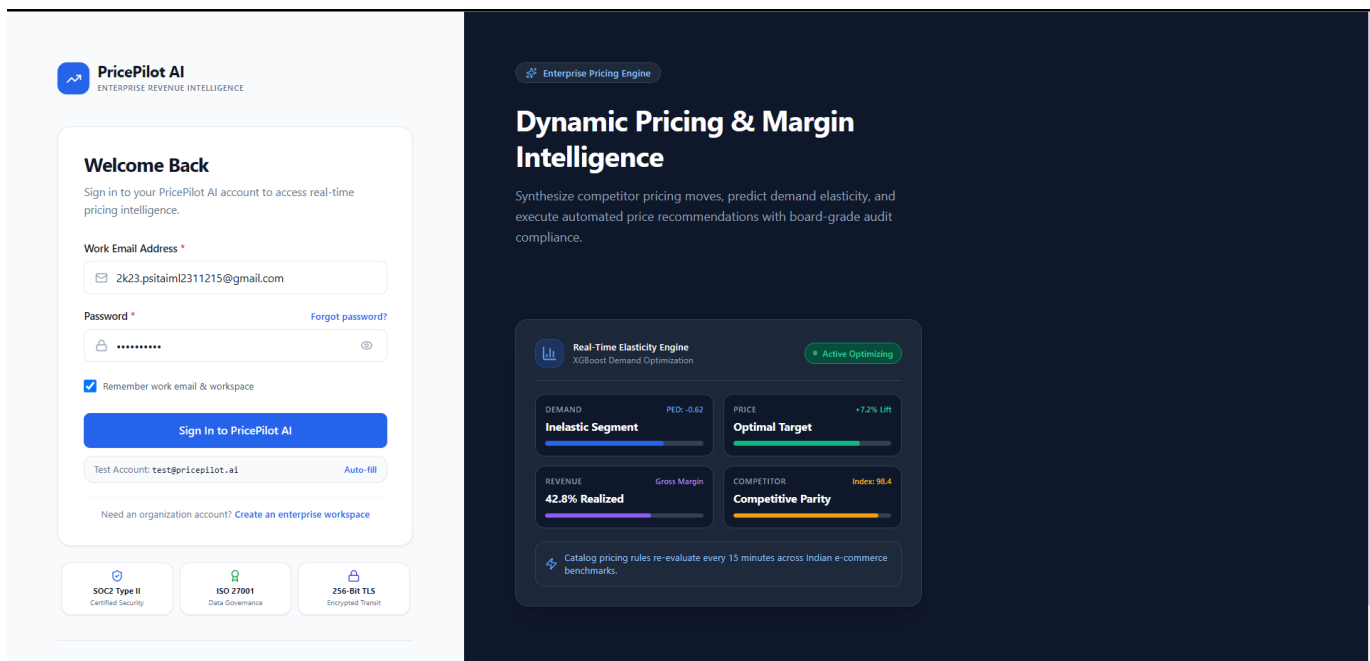
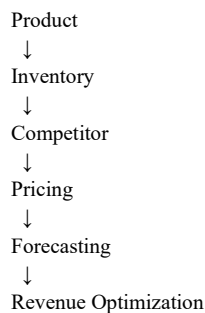
Product & Category Management

The Product module provides interfaces for product name, SKU, category, brand, cost price, selling price and status. Categories provide an additional dimension for product grouping and analytics.

Inventory Management

The Inventory module provides visibility into product availability and stock-related information. Inventory context is also important for pricing decisions because price strategy should consider product availability.

Core Business Flow



Step 4: Competitor Analysis & Market Intelligence

Competitor Analysis is one of the primary Milestone 3 modules. It provides a structured interface for managing competitors and comparing competitor pricing with the organization's products.

Competitor Management

- Add competitor
- View competitor list
- Update competitor information
- Match competitor products
- Record competitor prices
- View competitor price information

Competitor Price Comparison

The comparison interface provides market context by displaying the organization's product price alongside competitor prices. The comparison can be used to understand price differences, market benchmarks and relative price positioning.

Market Intelligence

Market Intelligence consolidates competitor and pricing information into a broader analytical view. It helps users understand competitive gaps and pricing position without manually comparing multiple screens.

Workflow



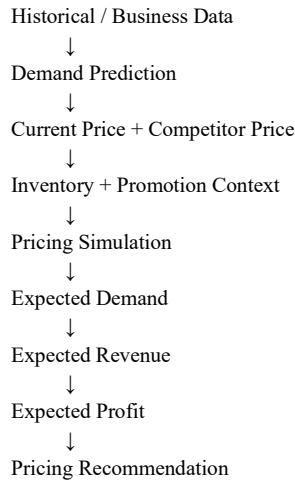
The screenshot displays the PricePilot AI web application interface. The browser address bar shows the URL `localhost:3000/products`. The application header includes the PricePilot AI logo, a user profile for Aditya Raj Pandey, and a search bar. The left sidebar contains navigation links for Overview, Catalog, Intelligence, and Reports. The main content area is titled 'Products' and shows a table of product data.

PRODUCT	SKU	CATEGORY	BRAND	COST PRICE	BASE PRICE	STATUS	ACTIONS
hp laptops HP AI powered Laptop	HP-1234	Electronics	HP	₹60,000	₹55,000	Active	View Edit Delete
Samsung Galaxy A55 Mid-range smartphone with AMOLED...	SAW-A55-128GB		Samsung	₹24,500	₹32,999	Active	View Edit Delete
Reebok Running Shoes Lightweight running shoes with breathable...	RBK-RUN-10001		Reebok	₹2,800	₹4,999	Active	View Edit Delete

Step 5: Pricing Intelligence & Price Simulation

Pricing Intelligence connects demand prediction, current pricing, competitor pricing, inventory and promotion information into a business-facing pricing workflow.

Pricing Intelligence Workflow



Price–Demand Simulation

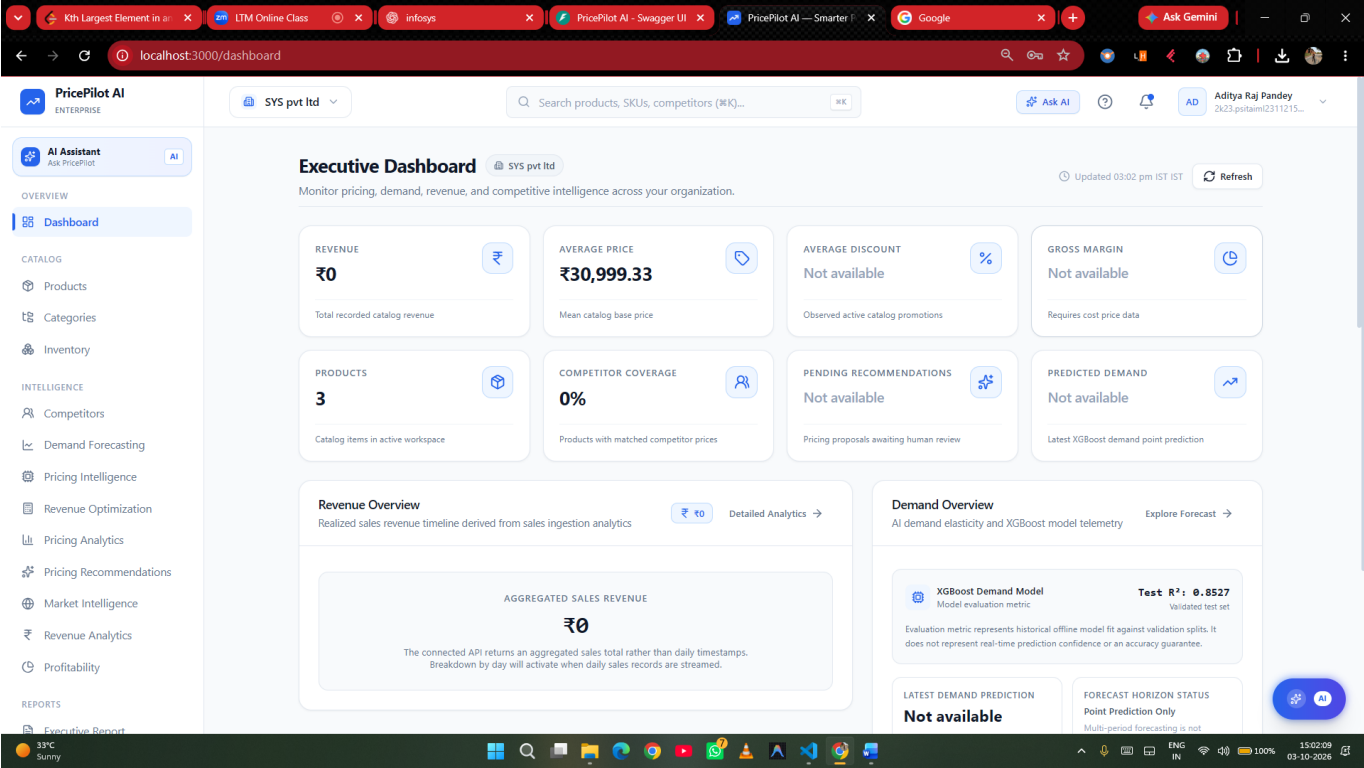
The pricing simulation interface supports scenario-based analysis rather than relying only on a single price output. Users can examine the expected effect of different candidate prices.

- Current price scenario
- Lower price scenario
- Higher price scenario
- Competitor benchmark scenario
- Custom candidate price scenario

Scenario Metrics

- Expected demand
- Expected revenue
- Expected gross profit
- Expected gross margin
- Competitor price position

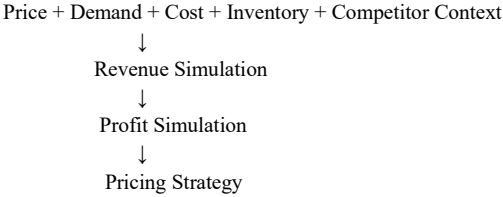
Historical actual values, model-predicted values and simulated values are kept conceptually separate so users do not confuse an estimate with an observed historical result.



Step 6: Revenue Optimization & Profitability Analytics

Revenue Optimization extends pricing analysis into business outcome analysis. The frontend provides a scenario-oriented interface for examining the relationship between price, demand, revenue, cost and profitability.

Revenue Optimization Workflow



Profitability Analytics

- Revenue
- Product cost
- Gross profit
- Gross margin
- Product profitability
- Category profitability

Financial Formatting

The India-focused frontend uses Indian Rupee formatting consistently across dashboards, tables, pricing screens and reports.

Example	Display
Small transaction	₹4,999
Business metric	₹1,25,000
Large business value	₹12,50,000
High-value metric	₹1,25,00,000

Business Interpretation

Revenue and profit are treated as related but distinct business measures. A price that changes revenue does not automatically produce the same change in profit because product cost and margin must also be considered.

Step 7: Forecasting & Pricing Analytics

The Forecasting frontend provides business-facing visualization for demand predictions and trend analysis. The project specification defines short-, medium- and long-term forecasting horizons.

Forecasting Horizons

Horizon	Periods	Business Use
Short-Term	7 / 14 / 30 days	Inventory planning, daily pricing, promotion planning
Medium-Term	3 / 6 months	Procurement, revenue forecasting, capacity planning
Long-Term	12 months	Strategic planning, product expansion, annual forecasting

Demand Trend

- Increasing Demand
- Stable Demand
- Decreasing Demand

Trend information is intended to be displayed alongside numerical demand predictions. Formal forecast confidence should only be presented as a calibrated measure when an appropriate confidence method has been implemented and validated.

Pricing Analytics

Pricing Analytics provides deeper visualization of pricing performance using interactive line charts, bar charts and scatter charts. These visualizations support revenue, demand, product, category and competitor analysis.

Visualization Examples

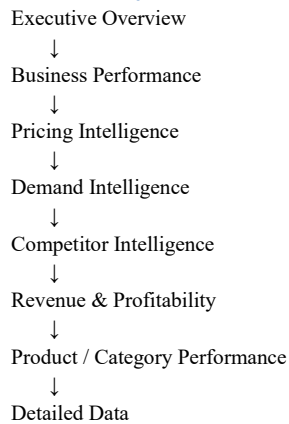
- Revenue trend line chart
- Demand trend chart
- Product comparison bar chart
- Category comparison bar chart
- Price-demand scatter chart
- Competitor comparison chart

The screenshot displays the PricePilot AI Enterprise web application. The browser address bar shows 'localhost:3000/pricing'. The application header includes the 'PricePilot AI ENTERPRISE' logo, a user profile for 'Aditya Raj Pandey', and a search bar. The left sidebar contains navigation links for Overview, Catalog, Intelligence, and Reports. The main content area is titled 'Pricing Intelligence' and 'Enterprise Decision Support'. It features a 'TARGET PRODUCT' section with a search bar and a list of products: 'Reebok Running Shoes' (SKU: RBK-RUN-10001, Price: ₹4,999), 'Samsung Galaxy A55' (SKU: SAM-A55-128GB, Price: ₹32,999), and 'hp laptops' (SKU: HP-1234, Price: ₹55,000). A toggle for 'XGBoost Model' with a test R² of 0.8527 is visible. The bottom of the screen shows a Windows taskbar with various icons and system information.

Step 8: Executive Business Intelligence Dashboard

The Executive Business Intelligence dashboard provides a management-oriented view of the system. It summarizes important business indicators before the user moves into detailed analytical modules.

BI Hierarchy



Executive KPI Areas

- Revenue performance
- Profitability
- Pricing performance
- Demand indicators
- Competitor position
- Product performance
- Category performance

Interactive Business Analysis

The executive dashboard is designed as an interactive analytical interface. KPI cards provide quick summaries, while charts and tables provide supporting detail.

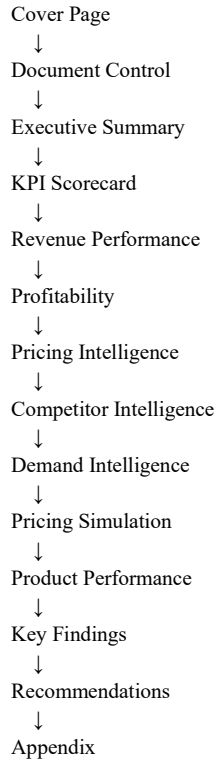
Design Principle

The dashboard is intended for exploration and decision support, whereas the formal report provides a structured management document. The two interfaces therefore serve different presentation purposes while using consistent underlying business data.

Step 9: Business Intelligence Reports

A structured business reporting workflow was implemented to convert application analytics into a formal report format.

Report Structure



Document Control

- Report ID
- Generated At
- Data As Of
- Reporting Period
- Organization
- Generated By
- Application / Report Version
- Timezone

Report Design

The report is designed as a formal business document rather than a screenshot export of the dashboard. It combines structured tables, KPI summaries, charts and analytical sections.

Consistency

The same financial formatting and business terminology used in the dashboard are applied to the report so users can move between interactive analysis and formal reporting without inconsistent presentation.

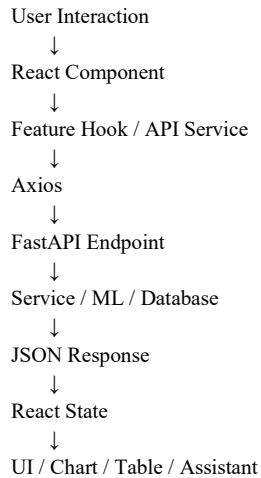
Step 10: AI Assistant & Frontend–Backend Integration

An AI Assistant interface was integrated into the frontend. The assistant communicates with backend AI services and displays the actual assistant output in a clean conversational interface.

AI Assistant Features

- Conversational interface
- Loading state
- Error handling
- Retry action
- Copy response
- Markdown rendering
- Clean extraction of assistant output

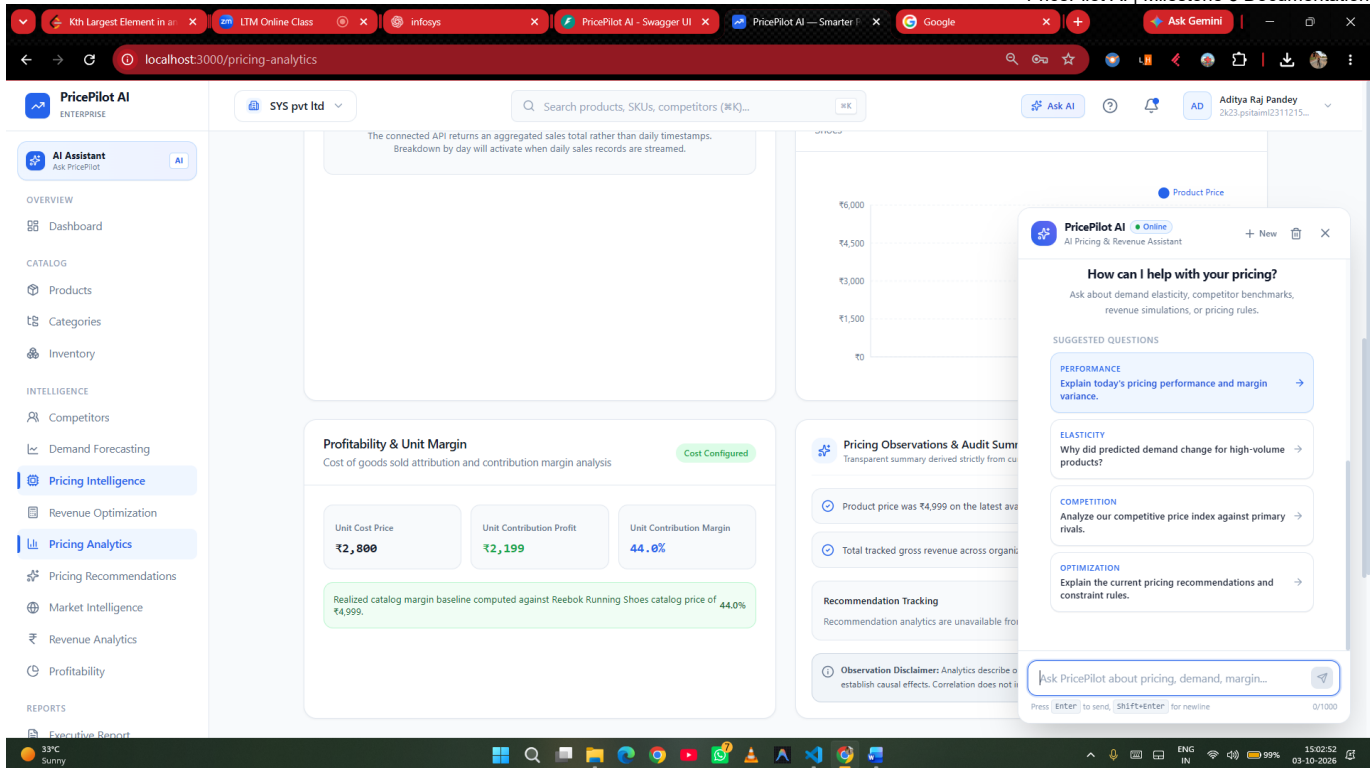
API Integration Architecture



Backend Integration Areas

- Authentication
- Users and organizations
- Products and categories
- Inventory
- Competitors
- Pricing
- Sales
- AI services

The frontend is therefore a presentation and interaction layer over the FastAPI application rather than a direct database client.



Step 11: Responsive Design, UX, Security & Quality

Responsive Design

- Desktop layouts
- Laptop layouts
- Tablet layouts
- Mobile layouts
- Responsive navigation
- Responsive charts and tables

User Experience

- Consistent typography
- Professional buttons
- Meaningful icons
- Reusable cards
- Clear information hierarchy
- Loading states
- Empty states
- Error states

Security Considerations

- Protected application routes
- Backend-controlled authorization
- Organization-aware API access
- Environment-based configuration
- No intentional exposure of backend secrets
- Frontend validation combined with authoritative backend validation

Error Handling

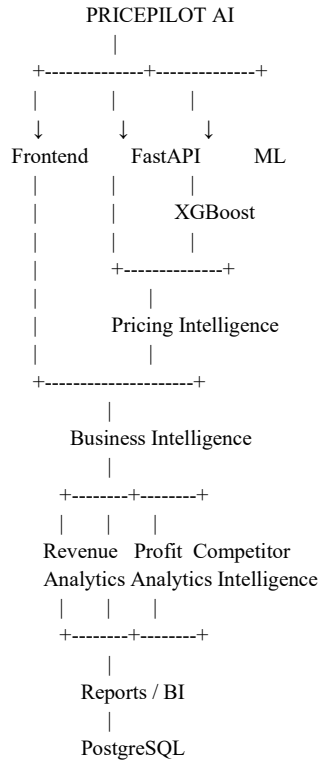
The frontend handles API failures, loading states, empty results and unexpected component errors. A React Error Boundary is used to prevent an isolated component failure from becoming an uncontrolled application failure.

Data Integrity

Predicted, simulated and historical values should be clearly distinguished. Financial values use centralized formatting to maintain consistency across the application.

Step 12: End-to-End Architecture, Testing & Challenges

End-to-End Architecture



Testing & QA Areas

- Route navigation
- Authentication flow
- Dashboard rendering
- API integration
- Product workflows
- Competitor workflows
- Pricing workflows
- Forecasting screens
- Revenue optimization
- Charts and tables
- Report generation
- AI Assistant
- Responsive layouts
- Error and loading states
- Production build validation

Challenges Faced

- Maintaining consistent frontend-backend API contracts.
- Connecting multiple business workflows instead of isolated pages.
- Presenting complex analytics through understandable visualizations.
- Separating actual, predicted and simulated information.
- Designing both interactive dashboards and formal reports.
- Maintaining responsive layouts across screen sizes.
- Handling API, rendering and data-state errors.
- Maintaining consistent Indian financial formatting.

Conclusion

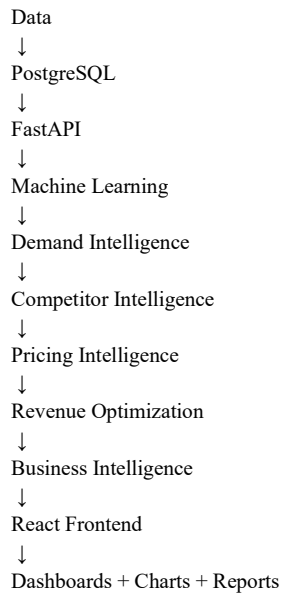
Milestone 3 extended PricePilot AI from a Machine Learning, backend and database foundation into a complete business-facing dynamic pricing and revenue intelligence application.

The milestone focused on competitor analysis, market intelligence, pricing simulation, revenue optimization, profitability analytics, executive business intelligence and frontend implementation.

Major Outcomes

- Competitor management and pricing comparison workflows.
- Market intelligence interface.
- Pricing intelligence and scenario simulation.
- Revenue optimization workflow.
- Profitability analytics.
- Pricing analytics dashboards.
- Demand forecasting interface.
- Executive business intelligence dashboard.
- Structured business report generation.
- AI Assistant interface.
- Responsive React frontend.
- Frontend–FastAPI integration.
- Interactive charts and reusable UI components.
- Authentication and protected application navigation.

Complete Business Workflow



The resulting frontend acts as the business-facing decision-support layer of PricePilot AI, connecting the project's Machine Learning, backend and database capabilities with interactive pricing, revenue and business intelligence workflows.

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